

Advanced Nmap Assessment of Linux Server

Professional Case Report 19

Author	Balsa Klikovac
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Case ID	CSC-2026-19
Discipline	Network assessment
Overall risk	Critical
Evidence status	Laboratory methodology supported by saved Nessus evidence

Purpose

Identify exposed Linux services and verify high-impact conditions without treating every open port as an equal risk.

Scope note

This report documents work performed in an authorized training environment. Confirmed findings are separated from methodology and expected laboratory behavior.

Executive Summary

Metasploitable 2 was assessed with advanced Linux-oriented discovery, service probing, NSE enumeration and custom packet controls. The Linux target exposed several independent routes to root access. Service retirement and platform replacement are more effective than patching isolated findings.

Case Profile

Scenario	Metasploitable 2 was assessed with advanced Linux-oriented discovery, service probing, NSE enumeration and custom packet controls.
Environment	Kali Linux against Metasploitable 2 on an isolated VMware network.
Tools	Nmap, NSE, NFS tools, FTP validation, Packet trace, Greppable output

Methodology

I used the following sequence to keep the analysis repeatable and to separate observation from interpretation.

1. Used broadcast discovery and Linux-specific Null and Xmas scan behavior.
2. Increased version-detection intensity and collected service banners.
3. Enumerated NFS and anonymous FTP exposure.
4. Checked vsftpd and distcc backdoor conditions.
5. Recorded packet traces, reasons and greppable output for repeatable analysis.

Technical Findings

ID	Finding	Risk	Evidence and analysis
F01	Unauthenticated bind shell	Critical	Saved Nessus evidence showed a shell on TCP 1524 that returned uid=0(root).
F02	Legacy remote services	Critical	rlogin, rsh, Telnet, NFS and vulnerable Samba significantly increased attack surface.
F03	Weak and obsolete cryptography	Critical	The host used unsupported Ubuntu 8.04 and Debian weak-key material.
F04	NSE exploitation checks	High	Targeted scripts can safely validate vsftpd and distcc conditions before full exploitation.

Representative Commands and Actions

The following commands or analysis actions represent the core of the practical workflow.

```
nmap -sN <target_ip>
nmap -sV --version-intensity 9 <target_ip>
```

```
nmap --script nfs-showmount,nfs-ls,nfs-statfs <target_ip>
nmap --script ftp-vsftpd-backdoor -p 21 <target_ip>
nmap --packet-trace <target_ip>
```

Risk Assessment

Risk	Level	Operational impact
Unauthenticated root access	Critical	Complete host compromise
Weak trust services	Critical	Remote access without modern authentication

Recommendations

1. Remove rlogin, rsh, Telnet and unauthenticated bind shells.
2. Upgrade the operating system and regenerate cryptographic keys.
3. Restrict NFS exports and FTP access.
4. Use targeted scripts to confirm remediation before rescanning broadly.

Knowledge Demonstrated

- Linux service enumeration
- NSE validation
- Packet tracing
- Risk-focused interpretation

Conclusion

The Linux target exposed several independent routes to root access. Service retirement and platform replacement are more effective than patching isolated findings.

Hands On Lab Practical 19

Completed as part of the authorized cyber security course practical work.